MODULE 1 · LECTURE 16

Decision Support Systems (DSS): Concepts, Components and Applications

How computer systems help farmers and managers make better decisions — the decision process, the parts and types of a DSS, the decision-analysis and optimisation methods inside it, agricultural examples, and why many tools go unused.

Learning objectives

After this lecture you should be able to:

1. Explain the decision-making process and classify decisions as structured, semi-structured or unstructured.
2. Define a decision support system and distinguish it from a management information system and an expert system.
3. Describe the components and types of DSS.
4. Solve simple farm decisions with payoff tables, decision trees and expected monetary value.
5. Formulate and solve a small farm-planning problem by linear programming and interpret binding constraints and shadow prices.
6. Rank options with weighted multi-criteria scoring.
7. Describe agricultural DSS and the factors that decide whether they are used.

Scope: Lectures 11–15 introduced the models, data and maps that a DSS draws on. This lecture shows how they are combined to support decisions. Expert systems and soil information systems follow in Lecture 17, and AI-based decision making in Lectures 19–28.

1. Decisions in agriculture

Farmers and agricultural managers decide continually: which crop and variety to grow, when to sow, irrigate and spray, how much fertiliser to apply, whether to store or sell, which machine to buy. Simon (1960) described decision making in phases — **intelligence** (finding and defining the problem), **design** (developing and analysing options) and **choice** (selecting one) — followed in practice by implementation and monitoring.

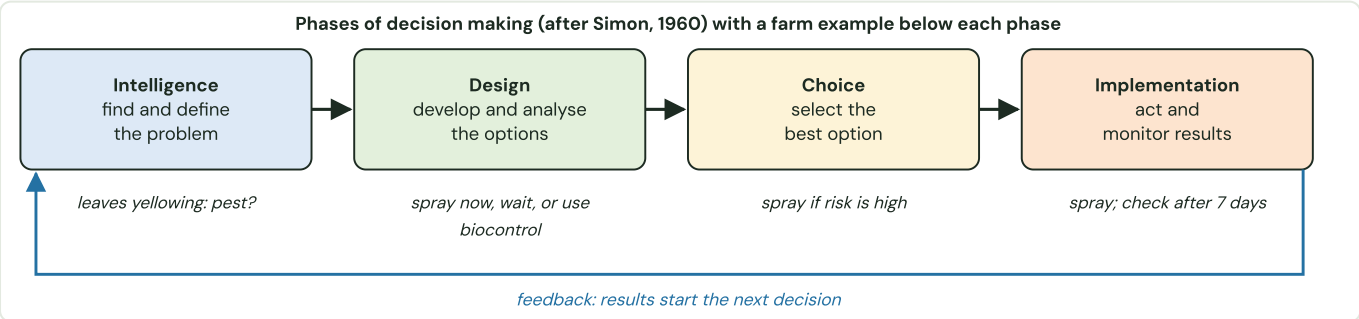


Figure 1. Phases of decision making with a farm example. Original diagram.

Type of decision	Description	Farm example
Structured (programmed)	Routine; the procedure and the right answer are known	Calculating the urea needed for a recommended N dose
Semi-structured	Part can be calculated, part needs judgement	Whether to spray now, given a pest forecast and weather
Unstructured (non-programmed)	New, complex, no set procedure	Whether to shift the whole farm to a new crop or enterprise

Decision support systems are most useful for **semi-structured** decisions: the computer handles data and calculations, and the person brings judgement and local knowledge.

2. What is a decision support system?

A **decision support system (DSS)** is an interactive computer system that helps users make semi-structured decisions by combining data, models and the user's own judgement. It supports, rather than replaces, the decision maker. Sprague (1980) proposed an early framework for DSS that described their technology levels, an evolving, iterative approach to building them, and the roles of the people who build and use them; the field has since grown to include data warehouses, web-based and group systems (Shim et al., 2002).

- **Interactive** — users change inputs and see results at once.
- **What-if (sensitivity) analysis** — explore how results change when prices, rainfall or other inputs change.
- **Goal seeking** — find the input needed to reach a target, e.g. the yield needed to break even.
- **Combines data and models** — databases, crop models, optimisation and statistics behind one interface.
- **Easy to use** — maps, charts and plain-language advice for non-specialists.

	Management information system (MIS)	Decision support system (DSS)	Expert system (ES)
Purpose	Routine reports on operations	Support semi-structured decisions	Give expert-like advice in a narrow field
Core	Database and standard reports	Data + models + interactive interface	Knowledge base of rules + inference engine
User role	Reads reports	Explores options, keeps control	Answers questions; receives a recommendation
Farm example	Monthly milk-collection report of a dairy cooperative	Crop-plan optimiser with what-if prices	Pest-diagnosis advisor (Lecture 17)

3. Components of a DSS

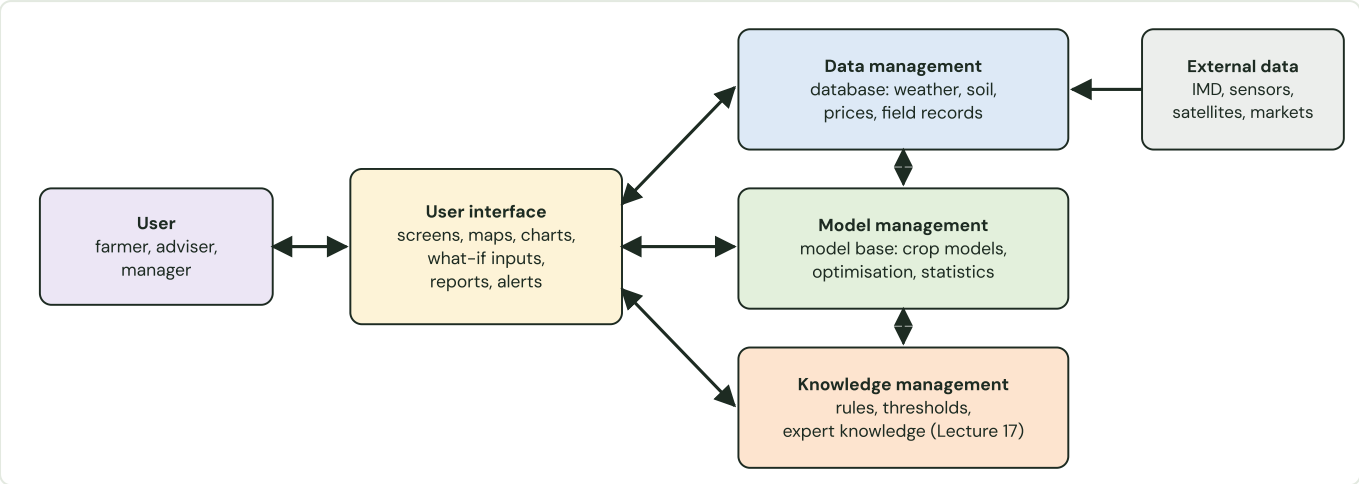


Figure 2. Components of a decision support system. Original diagram.

Component	Role	Agricultural content
Data management subsystem	Stores and retrieves data from internal and external sources (Lecture 6)	Weather, soil tests, field records, prices, satellite indices
Model management subsystem	Holds and runs models (the model base)	Crop simulation (Lecture 11), water balance (Lecture 13), linear programming, statistics
Knowledge management subsystem	Rules, thresholds and expert knowledge; makes the DSS "intelligent"	Economic thresholds for pests, fertiliser rules (Lecture 17)
User interface (dialogue)	How the user enters questions and sees results	Forms, maps, charts, mobile screens, alerts (Lecture 14)
User	Brings judgement and local knowledge; makes the final decision	Farmer, extension officer, FPO or scheme manager

4. Types of DSS

Type	Main emphasis	Agricultural example
Data-driven	Access to and analysis of large databases	Dashboards of mandi arrivals and prices; crop-area statistics
Model-driven	Simulation, optimisation and statistical models	Crop-model scenarios; farm-plan optimiser; irrigation scheduler
Knowledge-driven	Rules and expert knowledge	Pest and disease diagnosis advisors
Document-driven	Retrieval of documents and guidelines	Searchable packages of practices and advisories
Communication-driven (group)	Supporting shared decisions	Water-user associations planning canal rotations
Spatial (GIS-based)	Maps and spatial analysis (Lecture 15)	Site suitability for crops; watershed planning

The first five categories follow Power (2002); spatial DSS combine any of them with GIS.

5. Decision-analysis methods inside a DSS

5.1 Payoff tables and decision trees

When the outcome depends on an uncertain event, list the actions, the possible **states of nature** with their probabilities, and the payoff of each combination. The **expected monetary value (EMV)** of an action is the probability-weighted average of its payoffs.

$$\text{EMV}(\text{action}) = \Sigma\ [\ \text{probability of state} \times \text{payoff of the action in that state}\ ]$$

Worked example 1 — to spray or not? (illustrative values, per acre)
Crop value without damage ₹30,000. A pest outbreak (forecast probability 0.3) would cut yield by 30% if not sprayed; spraying costs ₹1,500 and limits the loss to 3%.
Payoffs: spray — outbreak 30,000 × 0.97 − 1,500 = ₹27,600; no outbreak ₹28,500. No spray — outbreak 30,000 × 0.70 = ₹21,000; no outbreak ₹30,000.
EMV(spray) = 0.3 × 27,600 + 0.7 × 28,500 = ₹28,230; EMV(no spray) = 0.3 × 21,000 + 0.7 × 30,000 = ₹27,300 → spray.
Break-even probability: spraying pays when 27,600p + 28,500(1 − p) > 21,000p + 30,000(1 − p), i.e. p > 1,500 ÷ 8,100 = 0.185. A pest-forecasting DSS supplies this probability.
Value of perfect information: with a perfect forecast the farmer would spray only when an outbreak comes: 0.3 × 27,600 + 0.7 × 30,000 = ₹29,280. EVPI = 29,280 − 28,230 = ₹1,050 — the most worth paying per acre for a perfect forecast.
Other criteria: maximin (best worst case) also chooses spray (27,600 > 21,000); maximax (best best case) chooses no spray (30,000).

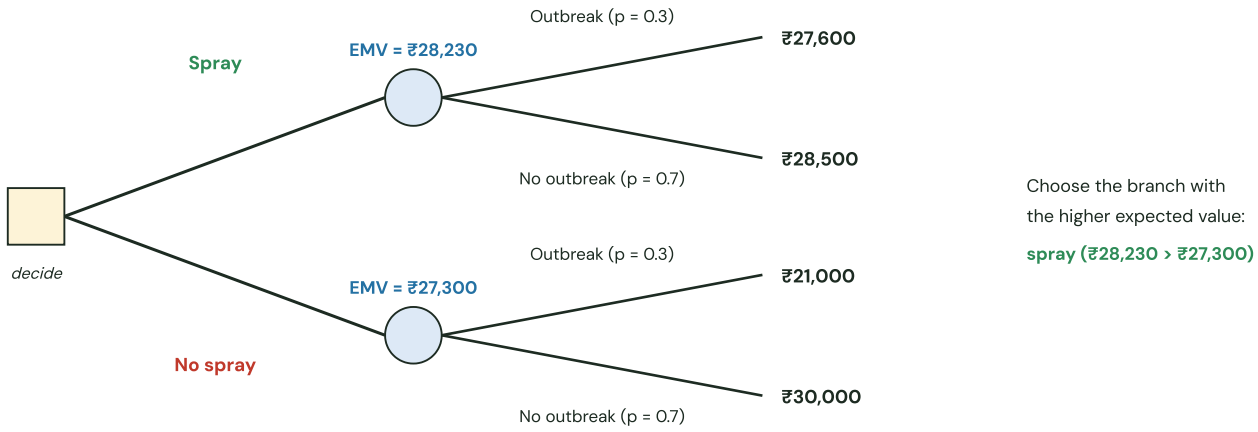


Figure 3. Decision tree for worked example 1: a square is a decision node and circles are chance nodes. Original diagram.

5.2 Linear programming

Linear programming (LP) finds the best allocation of limited resources — land, water, labour, capital — among competing activities. It has an **objective function** to maximise (profit) or minimise (cost), **decision variables** (hectares of each crop) and linear **constraints** (Taha, 2017). Farm-planning DSS use LP to suggest crop plans.

$$\text{Maximise } Z = 40{,}000x + 35{,}000y \qquad \text{subject to } x + y \le 10;\ 4{,}500x + 2{,}000y \le 35{,}000;\ 30x + 25y \le 270;\ x, y \ge 0$$

Worked example 2 — the best crop plan (illustrative coefficients)
x = ha of wheat (profit ₹40,000/ha, water 4,500 m³/ha, labour 30 days/ha); y = ha of mustard (₹35,000/ha, 2,000 m³/ha, 25 days/ha). Available: 10 ha land, 35,000 m³ water, 270 labour-days.
Corner points of the feasible region and profit: (0, 10) → ₹3,50,000; (4, 6) → ₹3,70,000; (6.38, 3.14) → ₹3,65,238; (7.78, 0) → ₹3,11,111.
Optimum: **4 ha wheat and 6 ha mustard, profit ₹3,70,000.** Land (10 ha) and labour (270 days) are fully used — they are **binding** constraints; only 30,000 m³ of water is used, leaving 5,000 m³ spare.
Shadow price of labour: with 280 labour-days the optimum moves to 6 ha wheat and 4 ha mustard, profit ₹3,80,000 — ₹1,000 more per extra day. Extra water is worth nothing at present because water is not binding.



Figure 4. Graphical solution of worked example 2: the shaded feasible region, the three constraints and the optimum. Original chart.

5.3 Weighted multi-criteria scoring

Many choices involve several criteria that cannot all be put in rupees. Each option is scored on each criterion, the scores are multiplied by weights that reflect the decision maker's priorities, and the weighted scores are added.

Criterion (weight)	Drip	Sprinkler	Surface
Water saving (0.35)	9	7	3
Low cost (0.25)	4	6	9
Labour saving (0.20)	8	7	4
Suitability for the crop and soil (0.20)	7	8	8
Weighted score	**7.15**	**6.95**	**5.70**

Worked example 3 — choosing an irrigation method (illustrative scores out of 10)
Drip = 0.35 × 9 + 0.25 × 4 + 0.20 × 8 + 0.20 × 7 = 3.15 + 1.00 + 1.60 + 1.40 = 7.15; sprinkler = 6.95; surface = 5.70 → drip ranks first.
What-if: if cost were weighted 0.45 and water saving 0.15, drip would score 0.15 × 9 + 0.45 × 4 + 0.2 × 8 + 0.2 × 7 = 6.15, sprinkler 6.75 and surface 6.90, and surface irrigation would rank first — the answer depends on the user's priorities, which a DSS makes explicit.

6. Agricultural DSS

DSS or tool	Decisions supported
DSSAT	Simulates crop management strategies to appraise new crops, products and practices before adoption (DSSAT Foundation, n.d.; Lecture 11)
CROPWAT	Crop water requirements, irrigation schedules and scheme water supply (Food and Agriculture Organization of the United Nations [FAO], n.d.; Lecture 13)
Nutrient Expert	Field-specific fertiliser recommendations for cereals (Pampolino et al., 2012; Lecture 13)
Weather-based agromet advisories (Meghdoot)	Sowing, spraying, fertiliser, irrigation and animal vaccination timed to the weather forecast (Lectures 11 and 14)
Pest surveillance and diagnosis (NPSS)	Pest identification and control advice from photos (Lecture 14)
Prescription-map tools	Variable-rate seed and fertiliser rates by zone (Lectures 12 and 15)
Farm-planning and marketing tools	Crop mix, input budgets, when and where to sell (worked examples 1–3; Lecture 14)

A survey of DSS for "Agriculture 4.0" reviewed thirteen representative systems for agricultural mission planning, water-resources management, climate-change adaptation and food-waste control, and identified challenges for the next generation of systems that draw on large streams of farm data (Zhai et al., 2020).

Why many tools are not used

In a study of farmers and advisers in the UK, Rose et al. (2016) found that uptake of agricultural decision support tools was disappointingly low despite their number, and identified fifteen factors that influence use — including **usability**, **cost-effectiveness**, **performance**, **relevance to the user** and **compatibility with compliance requirements**.

> **Design lessons for agricultural DSS:** involve farmers and advisers from the start; keep inputs few and familiar; give clear, specific advice with the reasons; work on phones and in local languages; show the value in rupees; update data automatically; and let the user stay in control of the final decision.

Steps in building a DSS

1. Identify the decision, the users and what they need.
2. Collect and organise the data; select or build the models.
3. Design the interface with users; build a prototype.
4. Test against real decisions and outcomes; refine iteratively.
5. Deploy with training and support; keep data and models up to date.

7. Key takeaways

- Decisions pass through intelligence, design and choice (Simon, 1960); DSS help most with semi-structured decisions.
- A DSS is interactive and combines data, models and the user's judgement; it supports what-if analysis and goal seeking.
- Its components are data, model and knowledge management, the user interface and the user.

- Types include data-, model-, knowledge-, document- and communication-driven DSS (Power, 2002), and spatial DSS.
- EMV and decision trees handle risk; a break-even probability and the value of perfect information guide pest and weather decisions.
- Linear programming finds the best use of limited land, water and labour; binding constraints and shadow prices show what limits profit.
- Weighted scoring ranks options on several criteria and makes priorities explicit.
- DSSAT, CROPWAT, Nutrient Expert and advisory apps are agricultural DSS; usability, cost-effectiveness, performance and relevance decide whether they are used.

8. Examination questions

Short-answer questions (2 marks)

Q1. Define a decision support system.

An interactive computer system that helps users make semi-structured decisions by combining data, models and the user's judgement.

Q2. Name Simon's phases of decision making.

Intelligence, design and choice (followed by implementation).

Q3. Distinguish structured and semi-structured decisions.

Structured decisions follow a known procedure; semi-structured decisions combine calculation with judgement.

Q4. List the components of a DSS.

Data management, model management, knowledge management, user interface and the user.

Q5. Name the five types of DSS described by Power.

Data-driven, model-driven, knowledge-driven, document-driven and communication-driven.

Q6. Define expected monetary value.

The probability-weighted average of the payoffs of an action.

Q7. What is the expected value of perfect information?

The difference between the expected payoff with perfect information and the best EMV without it.

Q8. State the three parts of a linear programming model.

Objective function, decision variables and constraints.

Q9. What is a binding constraint?

A constraint whose resource is fully used at the optimum.

Q10. What is a shadow price?

The increase in the objective value from one more unit of a scarce resource.

Long-answer questions (5–10 marks) — answer outlines

Q1. Explain the decision-making process and the role of DSS, distinguishing DSS from MIS and expert systems.

Figure 1; types of decisions; DSS definition and characteristics; comparison table.

Q2. Describe the components of a DSS with a diagram.

Figure 2 and table — data, model and knowledge management, user interface, user, external data.

Q3. Explain the types of DSS with agricultural examples.

Types table; spatial DSS; examples.

Q4. Solve a farm decision under risk using a payoff table and decision tree.

Worked example 1 and Figure 3 — EMV, break-even probability, EVPI, maximin and maximax.

Q5. Formulate and solve a farm-planning problem by linear programming and interpret the solution.

Worked example 2 and Figure 4 — corner points, optimum, binding constraints, shadow price.

Q6. Describe agricultural DSS and discuss the factors affecting their adoption.

DSS table — DSSAT, CROPWAT, Nutrient Expert, advisories, NPSS; Zhai et al. survey; Rose et al. factors; design lessons; building steps.

9. Glossary

Term	Meaning
Binding constraint	A constraint fully used at the optimum
Chance node	Point in a decision tree where an uncertain event occurs
Constraint	A limit on resources or requirements in a model
Decision node	Point in a decision tree where the decision maker chooses
DSS	Decision support system
EMV	Expected monetary value
EVPI	Expected value of perfect information
Feasible region	All combinations of decision variables that satisfy every constraint
Goal seeking	Finding the input needed to reach a target output
Linear programming	Optimisation of a linear objective subject to linear constraints
Maximin	Choosing the action with the best worst-case payoff
MIS	Management information system
Model base	The collection of models in a DSS
Objective function	The quantity to be maximised or minimised
Payoff table	Table of outcomes for each action and state of nature
Semi-structured decision	Decision partly computable and partly requiring judgement
Shadow price	Value of one more unit of a scarce resource
What-if analysis	Studying how results change when inputs change

References

DSSAT Foundation. (n.d.). *About DSSAT*. Retrieved November 9, 2026, from https://dssat.net/about/

Food and Agriculture Organization of the United Nations. (n.d.). *CROPWAT*. Retrieved November 9, 2026, from https://www.fao.org/land-water/resources/tools/software/cropwat/en

Pampolino, M. F., Witt, C., Pasuquin, J. M., Johnston, A., & Fisher, M. J. (2012). Development approach and evaluation of the Nutrient Expert software for nutrient management in cereal crops. *Computers and Electronics in Agriculture, 88*, 103–110. https://doi.org/10.1016/j.compag.2012.07.007

Power, D. J. (2002). *Decision support systems: Concepts and resources for managers*. Quorum Books. ISBN 978-1-56720-497-1

Rose, D. C., Sutherland, W. J., Parker, C., Lobley, M., Winter, M., Morris, C., Twining, S., Ffoulkes, C., Amano, T., & Dicks, L. V. (2016). Decision support tools for agriculture: Towards effective design and delivery. *Agricultural Systems, 149*, 165–174. https://doi.org/10.1016/j.agsy.2016.09.009

Shim, J. P., Warkentin, M., Courtney, J. F., Power, D. J., Sharda, R., & Carlsson, C. (2002). Past, present, and future of decision support technology. *Decision Support Systems, 33*(2), 111–126. https://doi.org/10.1016/S0167-9236(01)00139-7

Simon, H. A. (1960). *The new science of management decision*. Harper & Brothers.

Sprague, R. H. (1980). A framework for the development of decision support systems. *MIS Quarterly, 4*(4), 1–26. https://doi.org/10.2307/248957

Taha, H. A. (2017). *Operations research: An introduction* (10th ed., Global ed.). Pearson. ISBN 978-1-292-16554-7

Zhai, Z., Martínez, J. F., Beltran, V., & Martínez, N. L. (2020). Decision support systems for agriculture 4.0: Survey and challenges. *Computers and Electronics in Agriculture, 170*, Article 105256. https://doi.org/10.1016/j.compag.2020.105256

All diagrams and charts are original. Values in the worked examples are illustrative, and every result shown was recalculated, including all corner points of the linear programme.

Practice MCQs

Q1. Decision support systems are most useful for

(a) fully structured decisions (b) semi-structured decisions
(c) decisions with no data (d) routine bookkeeping

Q2. In Simon's model, developing and analysing possible courses of action is the

(a) intelligence phase (b) design phase
(c) choice phase (d) implementation phase

Q3. The DSS component that stores and runs crop and optimisation models is the

(a) data management subsystem (b) model management subsystem
(c) user interface (d) operating system

Q4. A DSS that works mainly from rules and expert knowledge is

(a) data-driven (b) model-driven
(c) knowledge-driven (d) document-driven

Q5. Expected monetary value is calculated as

(a) the largest payoff (b) the average of all payoffs
(c) Σ (probability × payoff) (d) the smallest payoff

Q6. In worked example 1, spraying is better when the outbreak probability exceeds about

(a) 0.05 (b) 0.185
(c) 0.5 (d) 0.9

Q7. The expected value of perfect information is

(a) the cost of spraying (b) the most worth paying for a perfect forecast
(c) the probability of an outbreak (d) the maximin payoff

Q8. Linear programming is used to

(a) draw maps (b) find the best allocation of limited resources
(c) store data (d) forecast weather

Q9. A constraint whose resource is fully used at the optimum is

(a) binding (b) non-binding
(c) infeasible (d) redundant

Q10. The extra profit from one more unit of a scarce resource is its

(a) opportunity cost
(b) shadow price
(c) expected value
(d) weight

Q11. Examining how results change when prices or rainfall change is called

(a) goal seeking
(b) what-if analysis
(c) normalisation
(d) compilation

Q12. Rose et al. (2016) found that the uptake of agricultural decision support tools was

(a) very high
(b) disappointingly low
(c) compulsory
(d) unmeasured

Answers

Q1 — (b) semi-structured decisions
Q2 — (b) design phase
Q3 — (b) model management subsystem
Q4 — (c) knowledge-driven
Q5 — (c) Σ (probability × payoff)
Q6 — (b) 0.185
Q7 — (b) the most worth paying for a perfect forecast
Q8 — (b) find the best allocation of limited resources
Q9 — (a) binding
Q10 — (b) shadow price
Q11 — (b) what-if analysis
Q12 — (b) disappointingly low